

FastAPI CORS Issue Resolution SOP

Objective

Restore frontend-to-API connectivity when browser requests are blocked due to missing or incorrect CORS configuration.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T014
Issue	FastAPI CORS issue
Category	API
Service	FastAPI
Severity	Low
Status	Resolved
Description	Frontend unable to call API
Expected Resolution	Enable CORS middleware in FastAPI

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm browser console error and identify origin, method, and headers being blocked.
2. Check whether the issue occurs only in browser clients and not in curl or Postman.
3. Review FastAPI CORSMiddleware configuration for allowed origins, methods, headers, and credentials.
4. Validate environment-specific frontend domains and API gateway paths.
5. Check preflight OPTIONS response and required Access-Control headers.
6. Update allowed origins or middleware configuration through approved deployment process.
7. Retest frontend calls for GET, POST, and preflight OPTIONS requests.
8. Monitor API access logs and update ticket with corrected origins.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Frontend blocked	Origin not allowed	Add approved frontend origin
Preflight fails	OPTIONS not handled	Enable CORS middleware correctly
Credentials blocked	Wildcard origin with credentials	Use explicit origins
Works locally only	Environment domain mismatch	Update environment config

Best Practices

1. Avoid wildcard origins in production.
2. Store allowed origins in environment-specific config.
3. Test CORS using browser and OPTIONS requests.
4. Review CORS after frontend domain changes.

References

[FastAPI Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [API Gateway Logs](#) | [Backend Runbook](#)